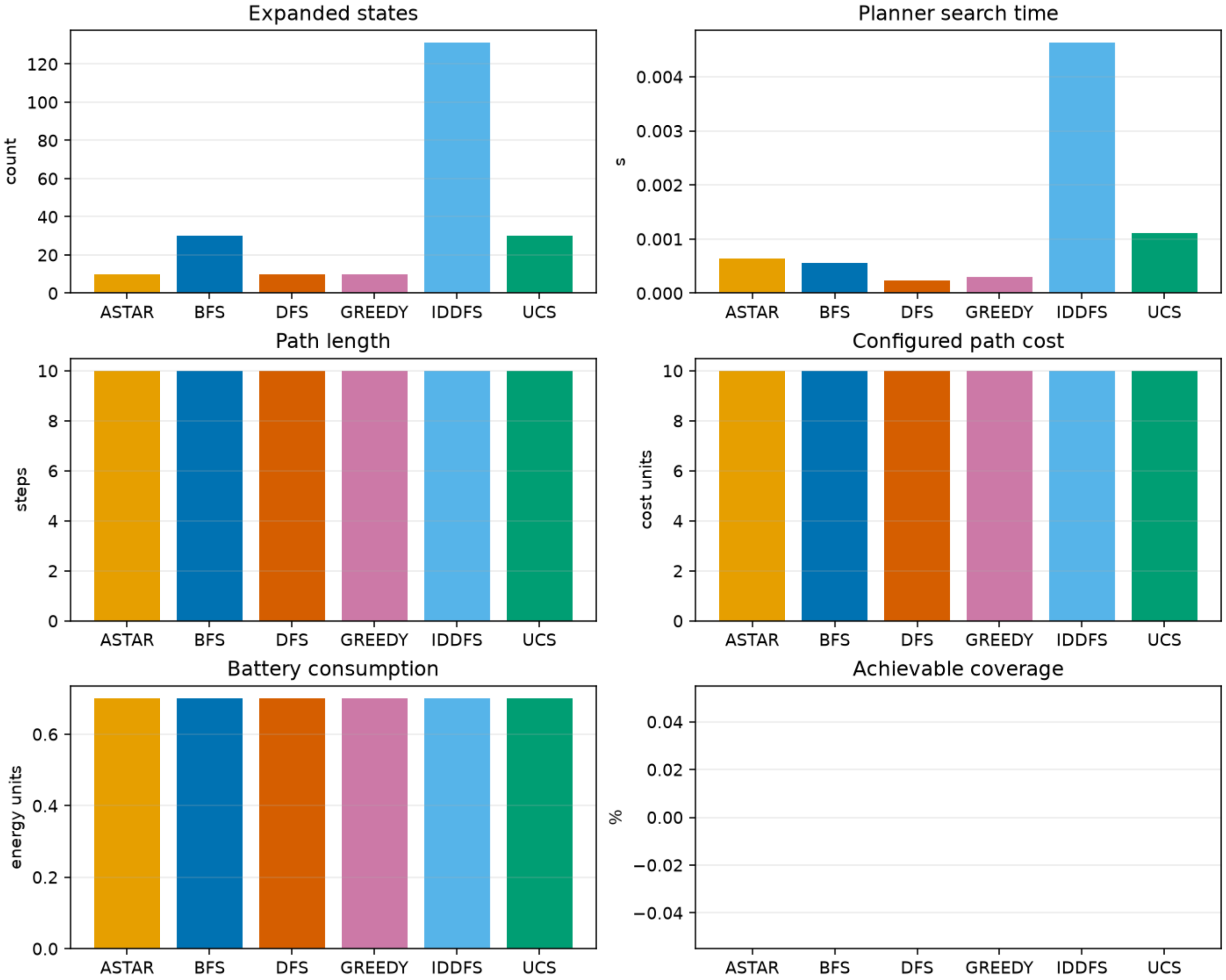
Point To Point comparison

Runs share map, robot, policy, neighbor order, cost model, heuristic, and mission mode.

Algorithm	n	Expanded	Search median (s)	IQR (s)	Steps	Cost	Energy	Coverage %	Stop
ASTAR	5	10.0	0.0006417999975[illegible]	[illegible]	[illegible]	[illegible]	0.69999999[illegible]9999318	NaN	mission_complete
BFS	5	30.0	0.00055540003811[illegible]	[illegible]	[illegible]	[illegible]	0.69999999[illegible]9999318	NaN	mission_complete
DFS	5	10.0	0.0002390999579[illegible]	[illegible]e-05	[illegible]	[illegible]	0.69999999[illegible]9999318	NaN	mission_complete
GREEDY	5	10.0	0.0002924000145[illegible]	[illegible]	[illegible]	[illegible]	0.69999999[illegible]9999318	NaN	mission_complete
IDDFS	5	131.0	0.0046236999332[illegible]	[illegible]	[illegible]	[illegible]	0.69999999[illegible]9999318	NaN	mission_complete
UCS	5	30.0	0.0011152999941[illegible]	[illegible]	[illegible]	[illegible]	0.69999999[illegible]9999318	NaN	mission_complete



Assumptions and limitations

Map grid: 6 × 6; resolution 1.0 m/cell. Movement: four-neighbor-cardinal; neighbor order: ['up', 'right', 'down', 'left']; heuristic: manhattan; cost model: {'transition': 'unit'}; tie-breaking: recorded algorithm default; coverage policy: None.

A single scenario does not support a generalized superiority claim. DFS depends on neighbor order. Greedy search is not guaranteed to return an optimal path. IDDFS may repeat expansions across depth limits. Simulated battery values are abstract model outputs, not real appliance measurements.